MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) –MG5DSECHE305 - MAIN GROUP ELEMENTS
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer any **10** questions
Each question carries **2** marks

1. Which alkali metal reacts least vigorously with water? Explain the reason [K] / [CO-1].
2. Name a suitable crown ethert that selectively complex with K+ ion [K] / [CO-1]
3. Sodium dissolves in liquid Ammonia to give blue colour solution because of …….. [A] / [CO-1]
4. Name one Calcium complex which is used in analytical applications. [K] / [CO-2]
5. State the inert pair effect and identify two heavy p-block elements that exhibit this phenomenon. [U] / [CO-2]
6. Describe graphite conduct electricity while diamond is an insulator. [U] / [CO-2]
7. Why are Zeolites known as molecular sieves? [U] / [CO -3]
8. How can you distinguish between boranes and carboranes using their composition? Give examples. [A] / [CO -3]
9. State the diagonal relationship? Give one pair of elements from Group 13 and Group 14 that exhibits it. [U] / [CO-3]
10. Point out any two important properties of halogens. [An] / [CO - 4]
11. Illustrate any one method for the preparation of XeF_2 and XeF_4. [An] / [CO - 4]
12. What are clathrates of noble gases? [An] / [CO - 4]

[10 x 2 = 20]

Part B

Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

13. Discuss the origin of the characteristic flame colours exhibited by S-block elements, giving suitable examples. [K] / [CO-1]
14. Describe the behavior of alkali metals dissolved in liquid ammonia. [K] / [CO-1]
15. Outline the structure of basic beryllium acetate with a labelled diagram. [U] / [CO-2]
16. Discuss the two structural modifications of Polymeric Boron Nitride. [U] / [CO-2]
17. Explain the structure and bonding in diborane using hybridization and molecular orbital concepts. [An] / [CO-3]
18. Determine the STYX code and structural formula for pentaborane-9 (B5H9). [A] / [CO-3]
19. Classify interhalogen compounds and explain the general methods of their preparation with suitable examples. [An] / [CO - 4]
20. Explain the VBT of structure and bonding of XeF_2 molecule. [An] / [CO - 4]

[6 x 5 = 30]

Part C

Short Essay Questions
Answer any **2** questions
Each question carries **10** marks

21. Discuss the diagonal relationship between lithium and magnesium, highlighting their similarities in chemical properties and supporting your answer with suitable examples. [K] / [CO-1]
22. Classifies the various allotropes of sulfur into distinct structural and thermodynamic categories based on their molecular geometry, phase stability, and ring strain characteristics. [An] / [CO-2]
23. Analyse the preparation methods and properties of silicon rubbers, and explain how their structure influences their characteristic properties. [An] / [CO-3]
24. Discuss the structures of ClF and ClF_3 type interhalogen compounds. Explain the preparation, structure, and reactivity with suitable reactions. [An] / [CO - 4]

[2 x 10 = 20]
